

A Support-Endpoint Counterexample to Conjecture 00000007737

Cosica

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Abstract

Conjecture 00000007737 asserts a formula for the lower endpoint x_k of the support of the k -fold free multiplicative convolution of the uniform law on $[0, 1]$. At $k = 2$ its formula gives $x_2 = 16e^{-2} > 1$. This is impossible: the free multiplicative convolution of two probability measures supported on $[0, 1]$ is itself supported on $[0, 1]$. Thus its lower endpoint is at most 1. This report gives the counterexample and an accompanying Lean 4 formalization of the endpoint contradiction.

1 The asserted formula

The conjecture defines $\mu_k = \nu \boxtimes \cdots \boxtimes \nu$, where ν is the uniform probability law on $[0, 1]$, and asserts

$$x_k = \exp(-2k + 2)k^{2k/(k-1)}.$$

There is no stated restriction excluding $k = 2$. Substitution gives

$$x_2 = \exp(-2)2^4 = \frac{16}{e^2}. \tag{1}$$

2 The support obstruction

Lemma 1. *If a and b are positive contractions in a C^* -algebra, then $a^{1/2}ba^{1/2}$ is a positive contraction.*

Proof. Positivity follows from

$$a^{1/2}ba^{1/2} = (b^{1/2}a^{1/2})^*(b^{1/2}a^{1/2}).$$

Since $0 \leq b \leq 1$, conjugation by $a^{1/2}$ gives

$$0 \leq a^{1/2}ba^{1/2} \leq a^{1/2}1a^{1/2} = a \leq 1.$$

□

Proposition 2. *The support of $\mu_2 = \nu \boxtimes \nu$ is contained in $[0, 1]$. Consequently its lower endpoint is at most 1.*

Proof. Use the operator definition of free multiplicative convolution (see, for example, the introduction of [1]): take two copies of the coordinate function $t \mapsto t$ in $C([0, 1])$, equipped with integration against ν , in their reduced free product probability space. The images a, b are free positive contractions with distribution ν ; positivity and the inequality $a, b \leq 1$ are preserved by the unital star homomorphisms. The distribution of $a^{1/2}ba^{1/2}$ is $\nu \boxtimes \nu$. Lemma 1 places the spectrum, hence the support of this distribution, in $[0, 1]$. A nonempty subset of $[0, 1]$ has infimum at most 1. □

3 Numerical contradiction

Lemma 3.

$$\frac{16}{e^2} > 1.$$

Proof. The elementary estimate $e < 3$ yields $e^2 < 9 < 16$. Dividing the strict inequality $e^2 < 16$ by the positive number e^2 proves the claim. $\square$

Combining Proposition 2, Lemma 3, and (1) gives $x_2 \leq 1 < x_2$, a contradiction. Hence the claimed formula, and therefore Conjecture 00000007737 as a conjunction containing that formula, is false.

4 Lean 4 verification

The included Lean project uses Lean 4.33.1 and Mathlib 4.33.1. It proves a stronger operator statement: for every pair of positive contractions in any nontrivial unital complex C^* -algebra, every scalar spectral probability law of their positive product has lower support endpoint different from the conjectured value at $k = 2$. Here a scalar spectral law is represented as a probability measure ρ on the real spectrum of $c = a^{1/2}ba^{1/2}$, pushed forward along the inclusion into $\mathbb{R}$. This is the standard representation of an operator's distribution in a state, obtained by restricting the state to the commutative algebra generated by c . The formal proof establishes:

- `positiveProduct_nonneg` and `positiveProduct_le_one` prove $0 \leq c \leq 1$;
- `positiveProduct_spectrum_subset` proves $\sigma_{\mathbb{R}}(c) \subseteq [0, 1]$;
- `spectralLaw_support_subset` and `spectralLaw_support_nonempty` prove the corresponding support bound and nonemptiness for the probability law;
- `claimedEndpoint_gt_one` proves $1 < 16/\exp(2)$, using Mathlib's certified bound $\exp(1) < 3$;
- `conjecturedEndpoint_two` checks the substitution into the original exponential and real-power formula;
- `conjecture_00000007737_endpoint_false` combines these facts for every such operator pair and spectral probability law.

The construction of free products and identification of their spectral laws with free multiplicative convolution are used at the level of the standard definition in the report; the Lean project does not implement a separate free-convolution operation. Its universal operator theorem needs neither freeness nor uniformity, so it applies to the defining realization above. In particular, the unit-support bound is proved, rather than assumed as a hypothesis of the final operator theorem.

The project introduces no additional axioms and uses no `sorry`, `admit`, or `native_decide`. The printed axiom dependencies are exactly `propext`, `Classical.choice`, and `Quot.sound`. Run `lake build` to check the proof and print these dependencies.

References

- [1] H. C. Ji, *Regularity Properties of Free Multiplicative Convolution on the Positive Line*, International Mathematics Research Notices 2021(6), 4522–4563. Preprint: <https://arxiv.org/abs/1903.02326>.